$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}}\left[\frac{\partial \hat{y}}{\partial o_{21}} \times \frac{\partial o_{21}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial w_{11}^1} + \frac{\partial \hat{y}}{\partial o_{22}} \times \frac{\partial o_{22}}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial w_{11}^1}\right]$$

↳ Very complex with only 2 hidden layers

Backpropagatn → chain diff (rule) + Memorization

↓ Maths ↓ CS (OP)

↳ Store the derivatives while calculated one time.

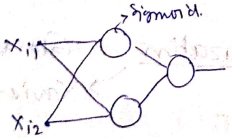
Vanishing Gradient Problem

In ML, the vanishing gradient problem is encountered when training ANNs with GD and backpropagation. In such methods, during each iteration of train each of the NN's weights receives an update proportional to the partial derivative of the error function w.r.t. the current weight. In worst case, this may stop the training of NN.

1) $0.1 \times 0.1 \times 0.1 \times 0.1 = 0.0001$ → VGP.

2) Deep NN →

3) Sigmoid/tanh → Activatn fn →

Back prop. 

$\hat{y} - y = L$

$w_\eta = w_0 - \eta\left\{\frac{\partial L}{\partial w_0}\right\}$ → derivative of the L wrt w

$$\frac{\partial L}{\partial w_{11}^1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial w_{11}^1}$$

↑ $0<x<1$ ↑ $0<x<1$

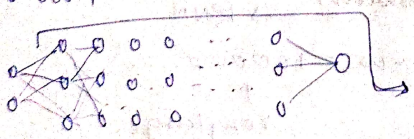
η ↑

$w_\eta = 1 - 0.01 \times 0.0001$

$w_\eta = 0.99999$

↳ did not changed

0.0001



too many layers.

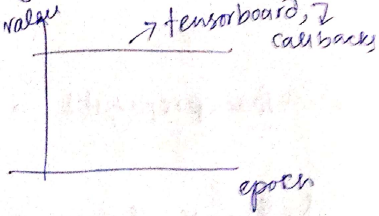
$\frac{\partial L}{\partial w_{11}^1} = 0.000000001$

↳ Stop training.

How to recognize VGP?

1) Loss focus → epoch → no changes → VGP

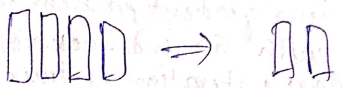
2) weights → graph → tensorboard, callbacks



keras → loss after every epoch

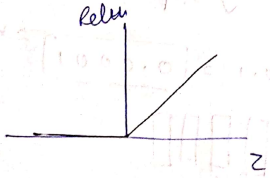
How to handle Vanishing Gradient Problem?

1) Reduce model complexity

 → problem cannot find complex patterns

2) Using ReLU Activation functions

max (0, z)



dying relu → derivative $w_\eta = w_0$

{leaky relu}

3) Proper weight initialization → Glorot / Xavier.

4) Batch normalization → Layer.

5) Residual Network → CNN → ResNET

↳ Building Block → ANN

Exploding Gradient Problem. → RNN

$w \times 10 \times 10 \ldots \times w = 1000000 \ldots$

↑ ↑ ↑ ↑

>1 >1 >1 very large.

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